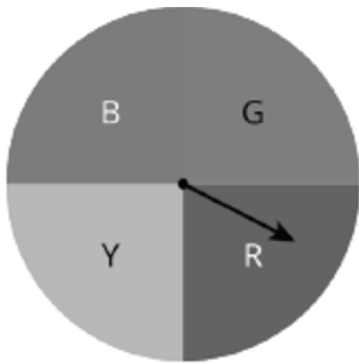


Grade 6 Accelerated
Unit 15
Lesson 5 Homework

Name _____
Date _____
Period _____

1. Explain the difference between theoretical and experimental probability of a simple experiment.

Tyriek performed a simulation and spun the arrow on the spinner below. He recorded his results in the frequency table below.



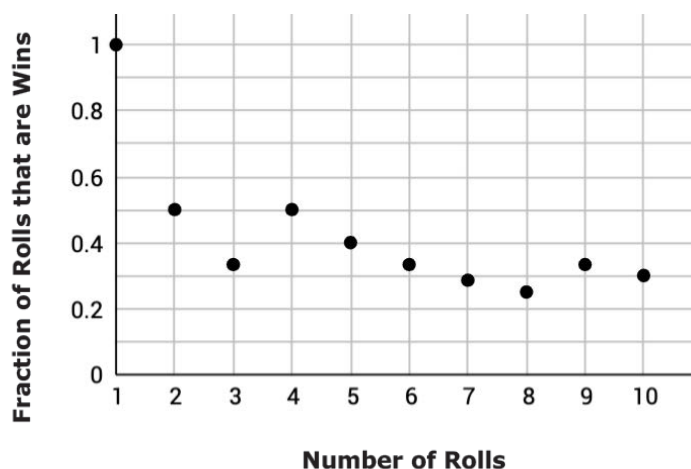
Color	Frequency
Blue	7
Green	9
Red	4
Yellow	5

2. What is the theoretical probability of landing on red?
3. What is the experimental probability of landing on red?

Arleni is playing a game where she wins if she rolls a 5 or a 6 on a standard number cube.

4. What is the theoretical probability that Alreni wins the game?

She then performs the simulation 10 times. The graph below shows the fraction of games that are wins.



5. Estimate the probability of a roll winning this game based on the graph.
6. What type of probability is this?
7. Draw a horizontal line on the graph representing the theoretical probability of rolling a 5 or a 6.
8. After 10 rolls, what fraction of the total rolls were a win?
9. If Arleni rolled the number cube 10 more times, estimate the probability that she will roll a 5 or a 6.
10. If she were to roll the number cube 500, how many times would you expect for her to roll a 5 or a 6